



ITM SKILLS
UNIVERSITY

MicroVest

A Fractional Stock Investment Simulation Platform

PS-7

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The Learning Gap

Students master investment theory in classrooms but rarely experience real market dynamics. Traditional stock prices create insurmountable barriers to practical learning.

Theory vs Practice

Finance courses teach concepts, but students need hands-on experience with portfolio management and market volatility.

High Entry Costs

Indian stocks like TCS (₹3,940) and Reliance (₹2,850) require substantial capital. A basic diversified portfolio demands ₹15,000–₹20,000.

Financial Risk

Students cannot afford to experiment with real money when learning fundamental investment strategies.



Price Comparison: Real vs Simulation

₹15K

Typical Investment

Minimum capital needed for real stock
portfolio

₹10

Simulation Entry

Starting amount for practice investing

1,500×

Lower Barrier

Reduced cost enables student
experimentation

The MicroVest Solution

A browser-based simulation platform that bridges theory and practice through fractional investing simulation.



Fractional Investing

Start investing from as little as ₹10. Practice with fractional amounts that simulate real market mechanics without financial risk.



Realistic Price Movement

Stock prices update every 2 seconds using Geometric Brownian Motion simulation. Track 10 simulated Indian stocks with authentic market fluctuations.



Portfolio Analytics

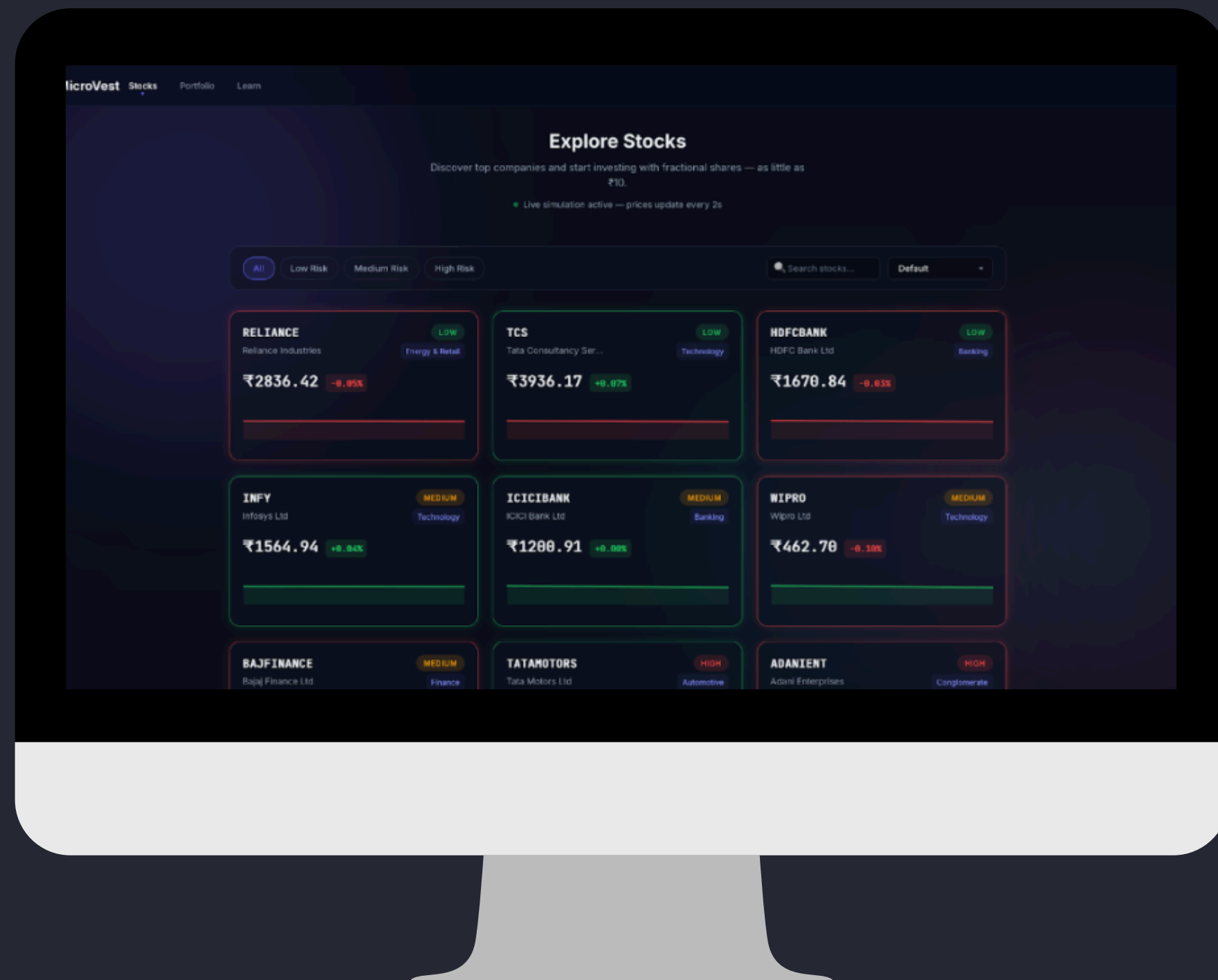
Real-time gain/loss calculations with visual charts. Monitor performance across multiple holdings with comprehensive tracking features.



Educational Module

Integrated glossary and quiz system. Learn financial concepts alongside practical application for reinforced understanding.

How MicroVest Works



Price Simulation Engine

Geometric Brownian Motion (GBM) algorithm generates realistic market fluctuations. Prices update every 2 seconds across 10 simulated Indian stocks.

Browser Interface

Users browse available stocks, invest fractional amounts, track portfolio performance, and access educational content—all through a responsive web interface.

Data Flow

Real-time updates delivered through Observer pattern architecture with LocalStorage persistence for portfolio data.

System Design Overview

1

User Interface

Browse stocks, invest, track portfolio

2

Price Simulator

GBM-based stock price generation

3

Portfolio Engine

Observer pattern for real-time updates

4

Data Storage

LocalStorage for persistence

User Journey Through MicroVest



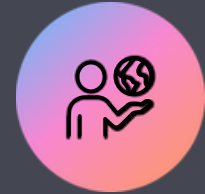
Made with

Platform Features in Action



Visual Charts

Inline SVG charts for price visualization across stocks



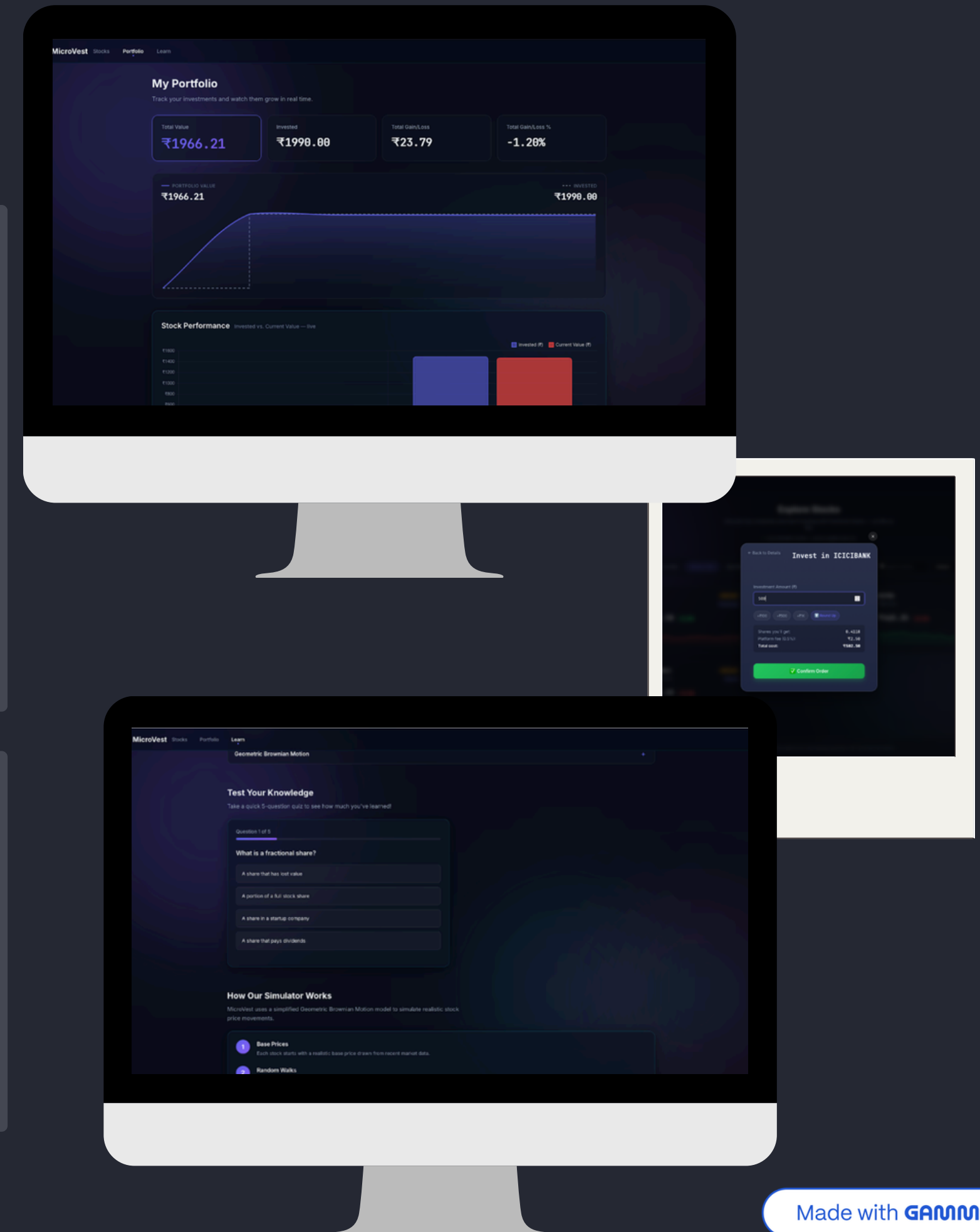
Responsive UI

Works seamlessly across desktop, tablet, and mobile devices



Glossary & Quiz

Educational module reinforces financial literacy





Thank You

MicroVest demonstrates how modern web technologies can create an interactive financial learning platform that helps students understand investing through practical simulation.